



Michael Beals, Project 4: AM Applications, ME127 Spring 2025



The “Nose What You Need”



Michael Beals
ME 127
Spring 2025



My Interpretation of the Prompt

This project explores the design of a daily-use fixture using the capabilities of additive manufacturing, particularly generative design, to consolidate frequently misplaced personal items into a single, functional, and playful structure.



Function

The primary function of the object is to consolidate key personal items—glasses, water bottle, keys, and airpods—into a single, easy-to-access location by the door. The design prevents tipping when placing a full water bottle at 70 degree angle and organizes frequently misplaced items. The intended user is myself, a student often leaving in a rush, who benefits from a centralized "grab-and-go" station that keeps everyday items visible and accessible. The structure also features four custom 3D-printed TPU noses, modeled from a scan of my own face, which serve as high-friction and soft mounting points for different styles of sunglasses.



Material Analysis

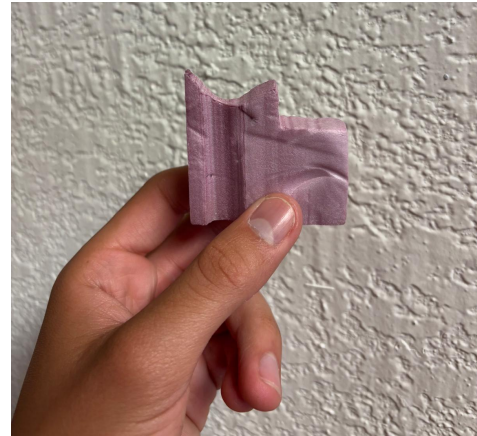
The base structure was printed in PLA due to its rigidity, affordability, and ease of use. Since stability was essential for resisting tipping—especially when holding a full water bottle—PLA was ideal for the structural elements. However, the sunglasses holders were printed in TPU, a flexible and grippy material. TPU was chosen for two main reasons:

1. **Texture and Flexibility:** TPU mimics the softness and squishiness of a real human nose, reducing the chance of damaging frames or deforming frames.
2. **Increased Friction:** Early PLA prototypes allowed some glasses to slide off, especially sport-style glasses. TPU's higher coefficient of friction improved grip, ensuring secure placement regardless of frame type.



My Journey

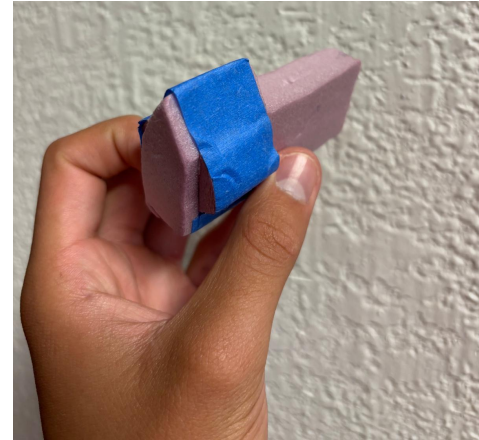
This is the original prototype I made that showcases my initial idea to make a “sunglasses tree” for the many sunglasses I have in my room. While this worked well with my regular sunglasses, it had some issues with my sports glasses, which were bulkier and weirdly shaped





My Journey

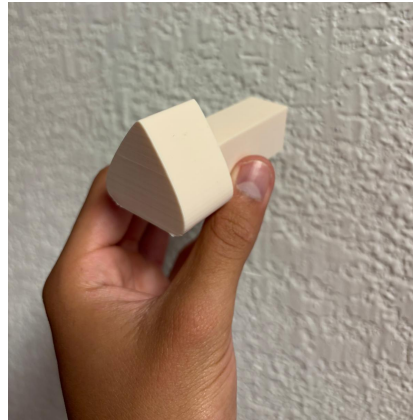
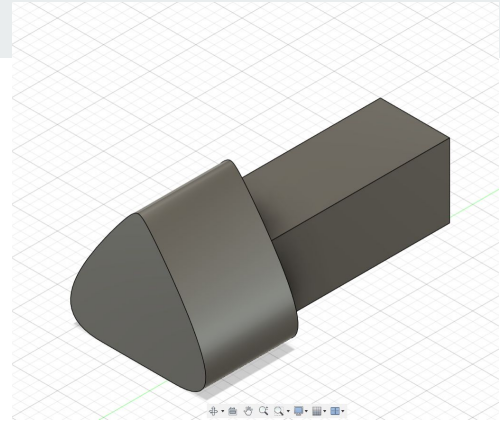
In my second prototype, I extended the spacing on the back, so that my sports sunglasses would fit. I additionally created a triangle-like shape that would mimic a nose and incorporate the padding on the glasses that already is designed to sit on something.





My Journey

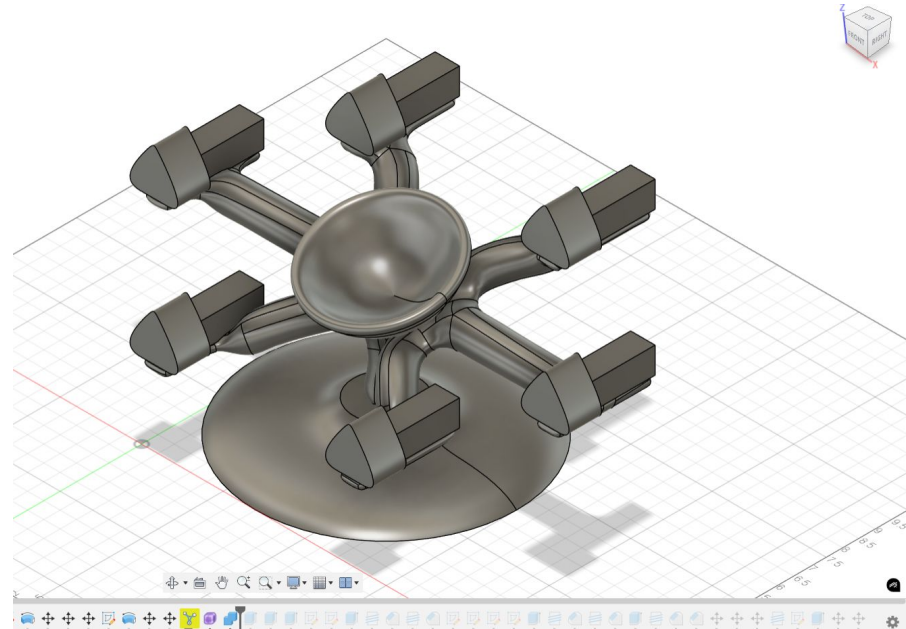
This is the Initial design I made based off the prototype.





My Journey

Before my milestone, I used automated modeling to make an initial “Glasses tree”. Last minute, I added a bowl to place keys into because I realized the place I put my sunglasses is the same place I put my keys.





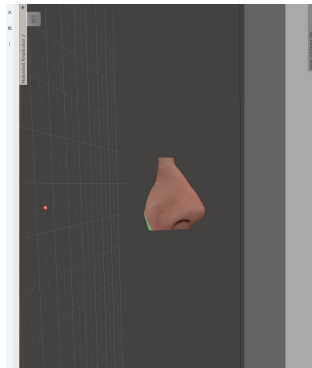
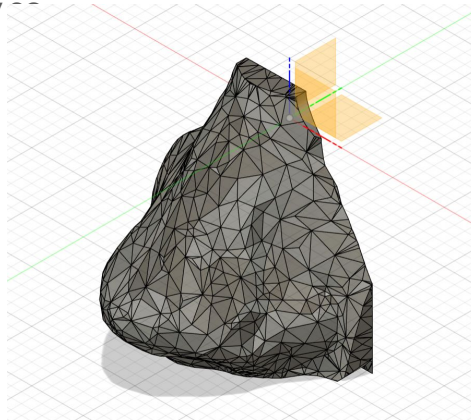
My Journey

Meeting with Joel! The meeting was very helpful. Thank you. He told me to consider the loads that would be going into this design, and gave me the idea to add my water bottle, as it would increase the size of the loads. Additionally, I asked him for advice on how to scan my nose and use that for the glasses holders.



My Journey

Scanning the nose: I used Luma 3D to scan my face/upper body. It gave me tremendous results. The level of detail was very high. Because I have a windows computer, I was able to use Meshmixer to cut out my nose and reduce the mesh complexity so that I could upload it to Fusion



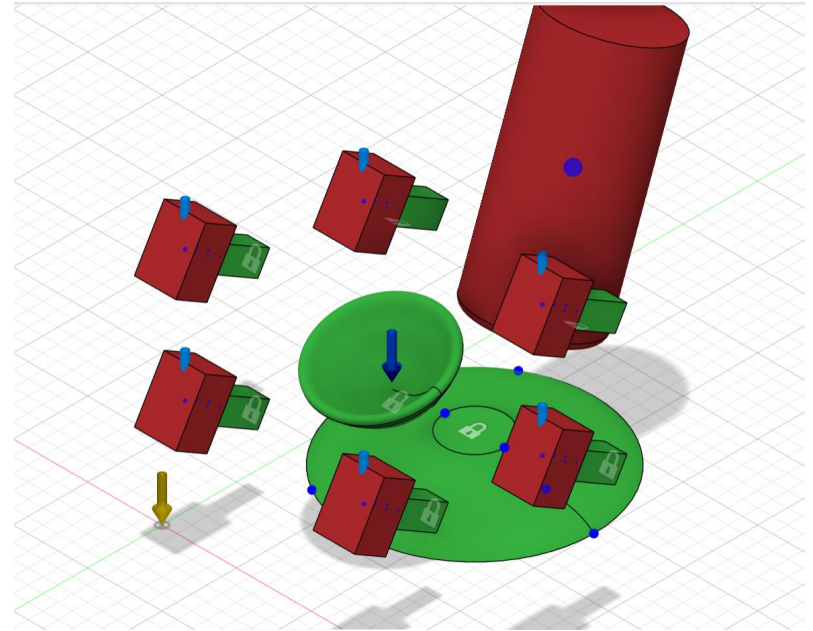
My Journey

I replaced the original mount with my nose and test printed it. It worked, but the glasses would slide a little down the nose. I decided that I would print out the nose separately out of TPU as it is softer and has higher friction, so the glasses wouldn't slide as much, and in the case of user misuse, the glasses wouldn't fall off as easily. I decided to use threading to connect the pieces. Test using pla:



My Journey

Now that everything was sorted, I decided to set up the model for generative design to create my structure, incorporating the added thermos. I used google and my own measurements to determine loads. In the case of the thermos and glasses, I used the maximums and designed for bigger/heavier thermoses so that it could be used for all kinds of glasses and bigger thermoses. After some initial designs, I found out that I had to create a very wide base to prevent tipping caused by the tilted thermos.



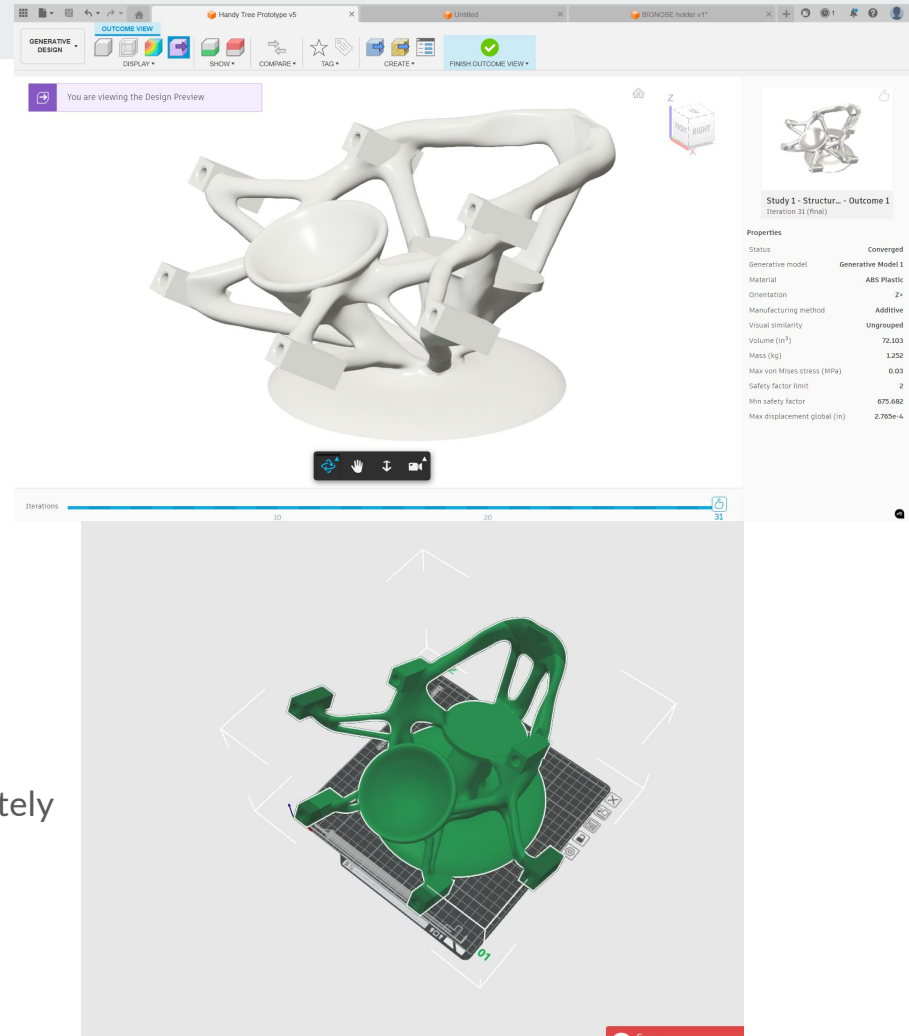
Initial setup: ^

My Journey

After a few tries and fails, redo of setup, and addition of misuse cases: I came up with the following design:

However, when I went to go print it, I was met with this:

The structure was far too big to print on the bambu printers. In my creativity, I had completely forgotten to keep into account the amount of space in the printer I had.

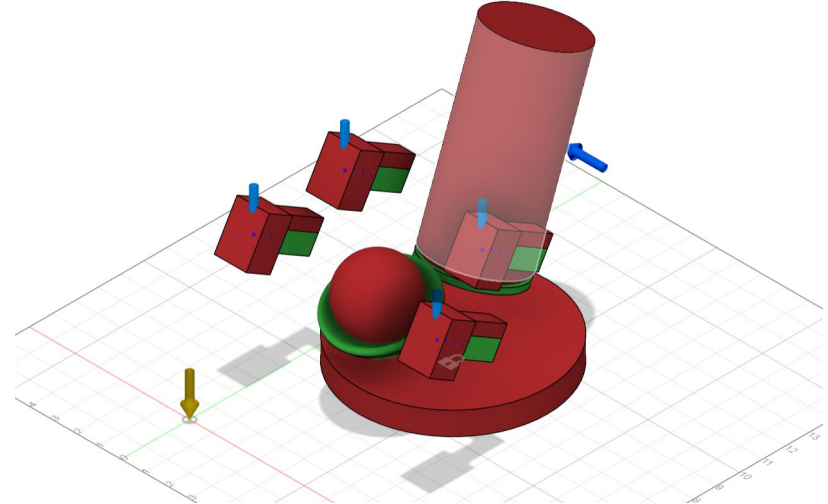
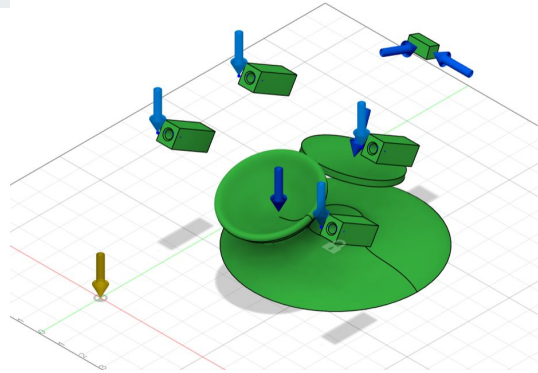
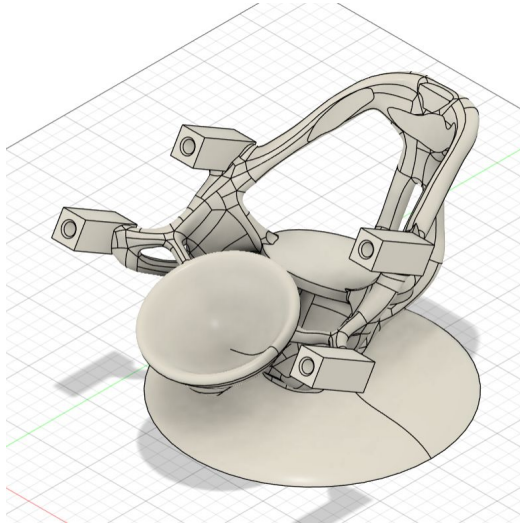


One of the Misuse cases:



My Journey

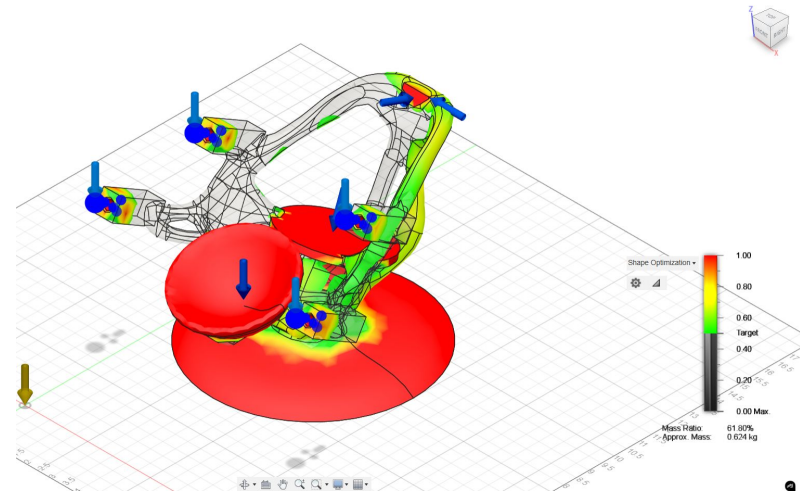
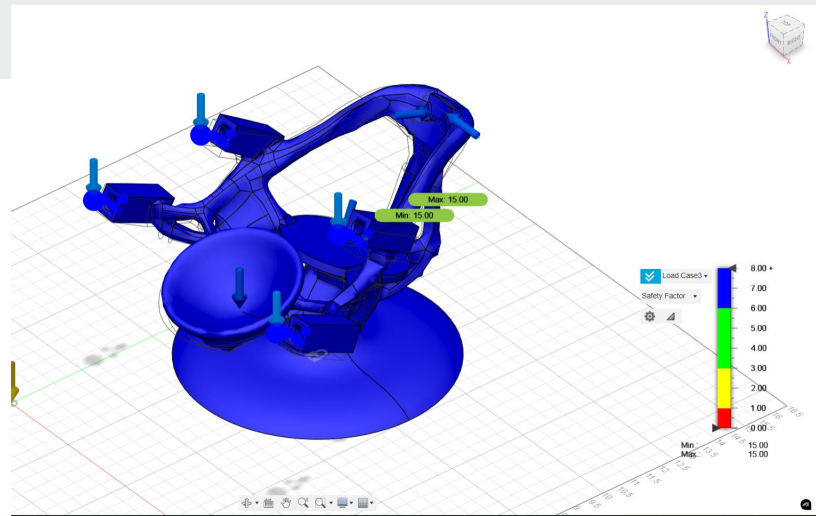
After a few more tweaks, shrinking down of the spacing, and removal of 2 ports for sunglasses: I got this:



My Journey

After getting my model, I ran a static stress analysis to test my result. The results were far better than I was expecting. The safety factor was over 15 everywhere. It was almost far too much. I decided to run a shape optimization using the same load cases and fusion gave me whatever this is:

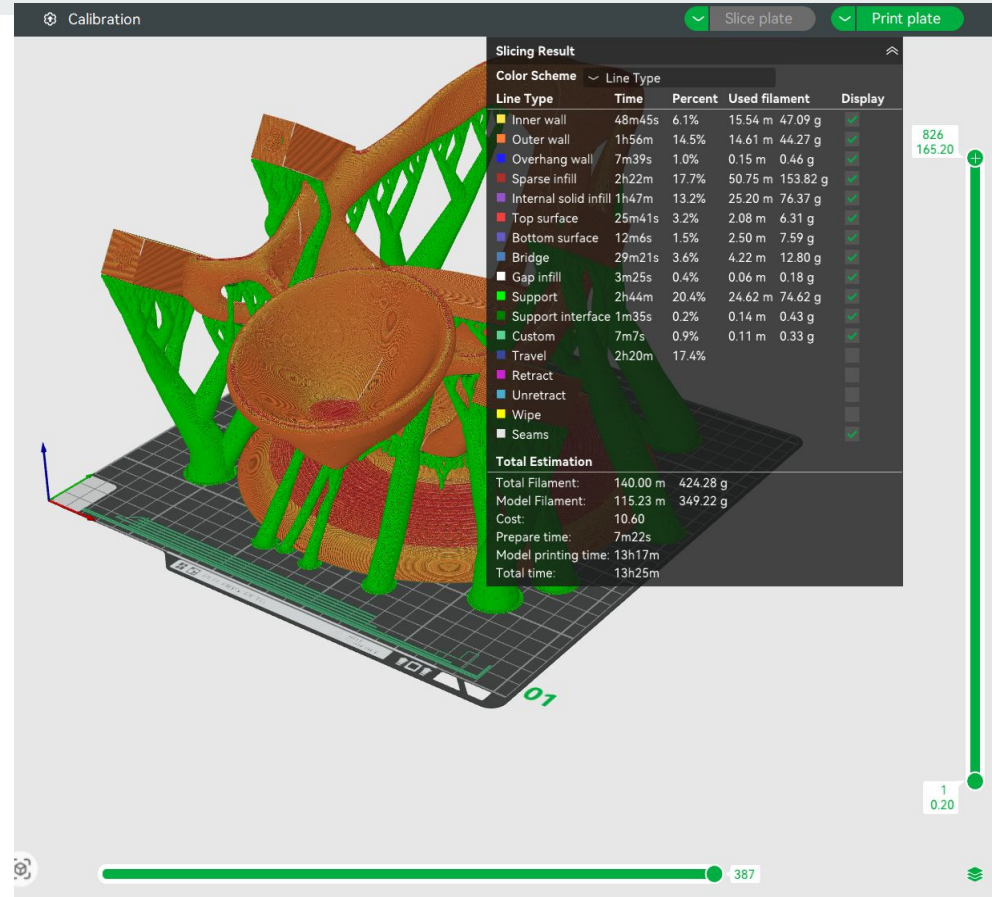
It cut off entire pieces. After some thought, I decided that since this was something I intended to use for myself, I would like to have something that would last and not fail from repeated stress/overuse. I decided to keep this model.



My Journey

My design barely fit into the print bed, and took 13 hours to print. Thankfully there were no issues with the print, and my first print at a full model would end up being my final.

Additionally I printed 4 TPU noses using the Ultimaker. Unfortunately I forgot to screenshot the sliced version. The print took 6 hours and cost 7 dollars. I printed on Ultimaker using standard TPU profile, 0.2mm layer height, and 20% infill. I also used breakaway supports. It worked, but just in case I had PLA noses that would work.





My Journey

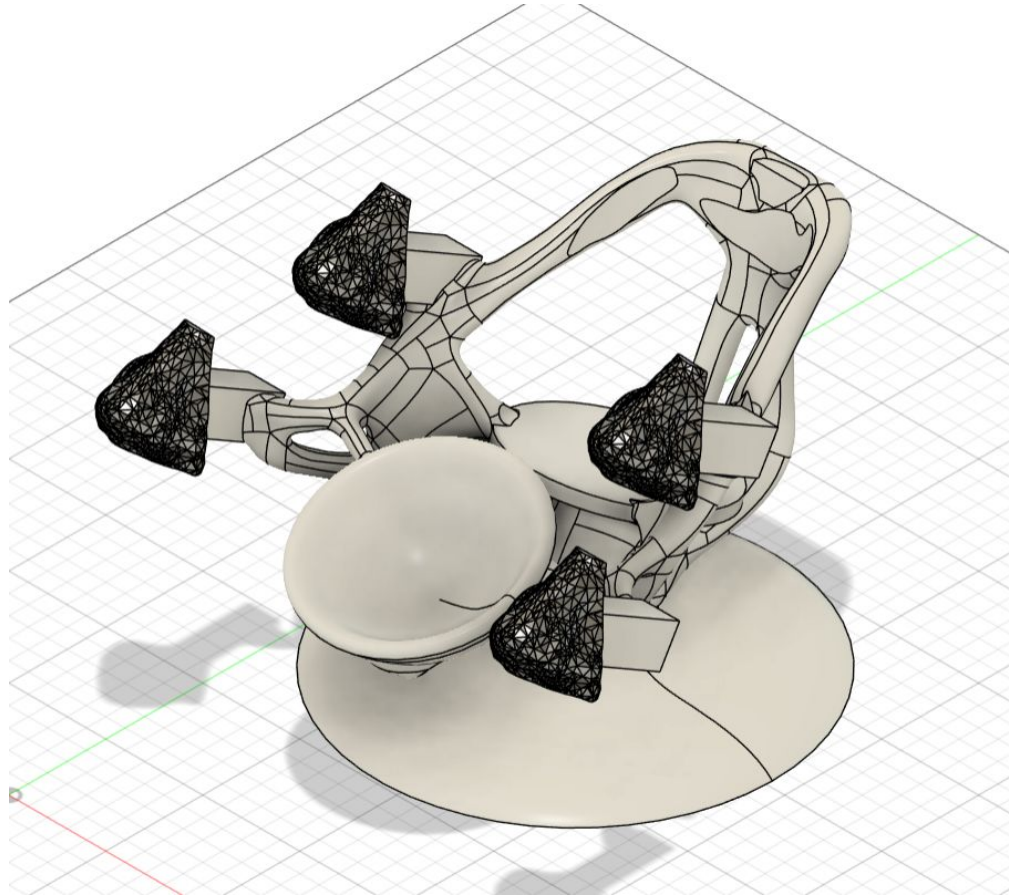
My final product:

Unfortunately, I picked up the TPU noses right before the presentations/taking these photos, so there some threading from the printer that I was unable to remove. I will try to remove as much as I can before meet the makers.





Final CAD Model:





Performance vs Intended Outcome

The final structure functioned exactly as intended. The PLA base resisted tipping even with a full water bottle placed toward the rear. Each TPU nose securely held a different style of glasses without slippage, and the tray provided a convenient spot for keys and earbuds. The design's playful character—made personal by using my own scanned nose—added delight without compromising function. Some drawbacks: because my initial design was too big for the printer, I had to shrink the design and get rid of two sunglasses holders. Doing so made the gap in front of the bowl for the keys/airpods a little small, forcing me to be cautious when grabbing the keys when sunglasses are present. Other than this, the performance was perfect.



Reflection:

This project helped me appreciate how additive manufacturing enables highly custom, low-volume functional parts that would be infeasible with traditional methods. I learned how to constrain generative design to resist unexpected forces and how critical small design choices (like nose angle and base radius) are in achieving stable, multi-use products. In future projects, I'll pay even more attention to how users physically interact with the object—especially when fun and function meet. In terms of this project: I like the model I have created, and have already started using it in my daily life. If I were to improve it, I would make it modular, so that I could add on to it as I go. That way, I could use multiple prints, be able to store more glasses, and have more spacing in between objects.



Final Costs:

Costs: All test prints were free. The structure cost \$10.60. The TPU noses and PLA noses cost 11 dollars combined. Therefore the total cost was around 22 dollars.

Time: I spent a decent amount of time iterating and reiterating the generative design and making changes as needed. Similarly scanning my nose and figuring out how to get the model into Fusion took longer than expected. In total, I believe it took me 20 hours (not including print time)